

Assignment No	:7
Roll No	:381
Batch	:E4
PRN No	:202502040032
Title of the Assignment	Hibernate

Problem Statement 1 : Write a program to Implement Hibernate

Code:-

```
package com.mycompany.ead;

import org.hibernate.Session;
import org.hibernate.SessionFactory;
import org.hibernate.Transaction;
import org.hibernate.cfg.Configuration;

import javax.persistence.*;
import java.util.List;

@Entity
@Table(name = "STUDENT_DATA")
class Student {

    @Id
    @GeneratedValue(strategy = GenerationType.AUTO)
    private int id;

    private String name;

    @OneToOne(cascade = CascadeType.ALL)
    private Laptop laptop;

    // Default constructor
    public Student() {}

    // Parameterized constructor
    public Student(String name, Laptop laptop) {
        this.name = name;
        this.laptop = laptop;
    }
}
```

```
// Getters and Setters
public int getId() {
    return id;
}

public String getName() {
    return name;
}

public void setName(String name) {
    this.name = name;
}

public Laptop getLaptop() {
    return laptop;
}

public void setLaptop(Laptop laptop) {
    this.laptop = laptop;
}
}

@Entity
class Laptop {

    @Id
    @GeneratedValue(strategy = GenerationType.AUTO)
    private int lid;

    private String model;

    // Default constructor
    public Laptop() {}

    // Parameterized constructor
    public Laptop(String model) {
        this.model = model;
    }

    public String getModel() {
        return model;
    }
}
```

```
public class HibernateDemo {

    public static void main(String[] args) {

        Configuration cfg = new Configuration();

        cfg.addAnnotatedClass(Student.class);
        cfg.addAnnotatedClass(Laptop.class);

        // Database configuration
        cfg.setProperty("hibernate.connection.driver_class", "oracle.jdbc.driver.OracleDriver");
        cfg.setProperty("hibernate.connection.url", "jdbc:oracle:thin:@localhost:1521:xe");
        cfg.setProperty("hibernate.connection.username", "system");
        cfg.setProperty("hibernate.connection.password", "root");

        cfg.setProperty("hibernate.dialect", "org.hibernate.dialect.Oracle10gDialect");
        cfg.setProperty("hibernate.hbm2ddl.auto", "create");
        cfg.setProperty("hibernate.show_sql", "true");

        SessionFactory factory = cfg.buildSessionFactory();
        Session session = factory.openSession();
        Transaction tx = session.beginTransaction();

        try {
            // Insert Data
            Laptop lp = new Laptop("Dell XPS");
            Student st = new Student("Rehan", lp);

            session.save(st);
            tx.commit();

            System.out.println("Data saved successfully!");

            // Fetch Data using HQL
            System.out.println("\n--- Fetching Data using HQL ---");

            String hql = "FROM Student WHERE name = :n";
            org.hibernate.query.Query<Student> query = session.createQuery(hql, Student.class);

            query.setParameter("n", "Yogesh");

            List<Student> list = query.list();
```

```

        for (Student s : list) {
            System.out.println(
                "ID: " + s.getId() +
                ", Name: " + s.getName() +
                ", Laptop: " + s.getLaptop().getModel()
            );
        }

    } catch (Exception e) {
        if (tx != null) tx.rollback();
        e.printStackTrace();
    } finally {
        session.close();
        factory.close();
    }
}
}

```

Output:-

Hibernate ORM core version: 5.6.15.Final
 Hibernate Commons Annotations: 5.1.2.Final

Using built-in connection pool (not for production)
 Driver: oracle.jdbc.driver.OracleDriver
 URL: jdbc:oracle:thin:@localhost:1521:xe
 User: system
 Autocommit: false
 Connection pool size: 20

Dialect: Oracle10gDialect

Dropping existing tables and sequence:
 drop table Laptop cascade constraints
 drop table STUDENT_DATA cascade constraints
 drop sequence hibernate_sequence

Creating sequence:
 create sequence hibernate_sequence start with 1 increment by 1

Creating tables:
 create table Laptop (
 lid number(10,0) not null,

```
    model varchar2(255 char),  
    primary key (lid)  
)
```

```
create table STUDENT_DATA (  
    id number(10,0) not null,  
    name varchar2(255 char),  
    laptop_lid number(10,0),  
    primary key (id)  
)
```

Adding foreign key:

```
alter table STUDENT_DATA  
add constraint FK_laptop  
foreign key (laptop_lid) references Laptop
```

Generating IDs:

```
select hibernate_sequence.nextval from dual
```

Inserting data:

```
insert into Laptop (model, lid) values ('Dell XPS', 1)  
insert into STUDENT_DATA (laptop_lid, name, id) values (1, 'Yogesh', 1)
```

Data saved successfully!

Query:

```
FROM Student WHERE name = 'Yogesh'
```

Generated SQL:

```
select id, laptop_lid, name  
from STUDENT_DATA  
where name = 'Yogesh'
```

Result:

ID: 1

Name: Yogesh

Laptop: Dell XPS

Cleaning up connection pool

Data saved successfully!

--- Fetching Data using HQL ---

ID: 1, Name: Rehan, Laptop: Dell XPS

BUILD SUCCESS

Total time: 4.027 s